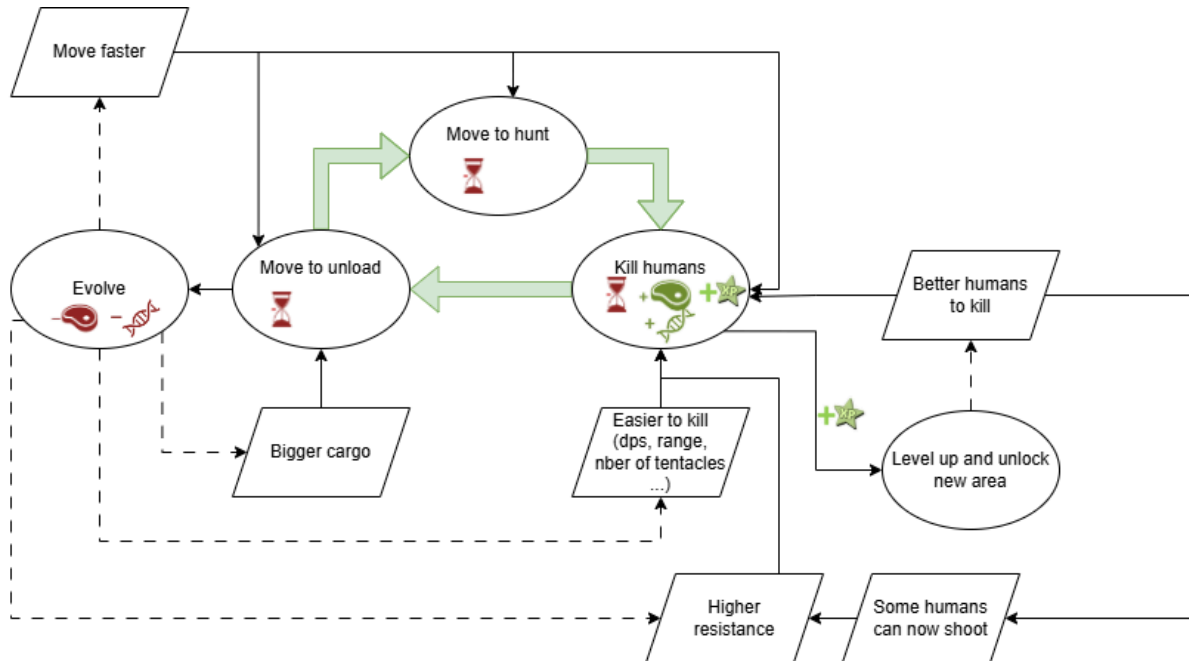


Game Analysis

Game Loop

Basic game loop

Here is the basic game loop of the game



I wanted to keep it simple, but to make it more accurate, let's add some additional info

Addition to Game Loop

- + DNA icon encompasses DNA itself and Eggs
- + DNA icon can only be found on specific humans
- DNA exists in several versions (one color for each) corresponding to different levels (green, then purple, then red). If a specific human type can drop DNA, it will drop DNA of a specific color
- Eggs can appear in the main evolution feature, but are mainly used in the secondary evolution feature called "Black market", which targets the same goals (Easier to kill, Higher resistance...) with existing improvements (speed, cargo, health...) or new ones (human cost, armor, slow...)
- There are at least 2 other secondary evolution systems:

- Farm, where you transform Time into Food. You can Invest Food to have a better Food-per-TimeUnit ratio online (Income) or offline (Capacity)¹
- Bots, which needs some Hard Currency to be initiated, and uses food for evolution. Catchers are a weaker but autonomous version of you, Deliverers help you but only focus on your cargo
- A more complex loop would encompass Quests, but these Quests can have too different behaviors (makes the loops fuzzier):
 - Most Quests have Specific Item as input but it can also be “finish another quest” or “level-up farm”
 - Finding Specific Items can require
 - To kill any target (Baby Steps)
 - To kill already existing specific targets (The Rescue)
 - To kill targets created by the Quest (Catch a satellite)
 - Output can be very different: an alien for Kustov Power main quest, Hard Currency (green gems), Soft Currency (food), XP...

Focus on time

By analyzing the game loop, it becomes obvious that the key resource is not soft-currency (food) but time:

- time to kill
- time to go back home and unload
- time to reach the next group of humans to kill, but also time to wait to have the interesting ones (usually the ones with specific resources) to respawn
- time transformed into soft currency by the farm

Choosing humans to kill

Killing humans may seem simple but it is the most complex activity, as finding a Target Group is the result of a multi-factor approach:

1. Which resource will I primarily kill for?
 - a. Are there resources that are not my main current goal, but need time to spawn and are currently available?
 - b. Are there resources that are not my main current goal, but are fast to get on the way?
2. What is the food-per-second ratio on this target?
3. What is the cargo-per-food ratio on this target?
4. Considering (2) and (3), what is the final food-per-second ratio, including the time needed to unload?
5. Can I stay still to hunt this Target?
 - a. If I can't, can I however hunt multiple Targets in the Group at once or should I choose one?
 - b. If I can, is my Deliverer big enough and quick enough to go almost idle? (works with big targets with a decent respawn rate)

¹ Capacity can be seen as offline Food-per-TimeUnit as it is almost always full when a new session starts (30 min max currently for me). So it corresponds to “Knowing I will come back in X hours and my farm will be full, how much food per hour have I generated?”

6. Is there a shooting Group nearby, how likely is it that it will attack me, and can I survive long enough to kill my Target?

As a consequence, unlocking a new area and “higher” Target Groups doesn’t necessarily mean changing Target.

Monetization

Monetization models

The game uses both Ad-monetization and In-app purchases.

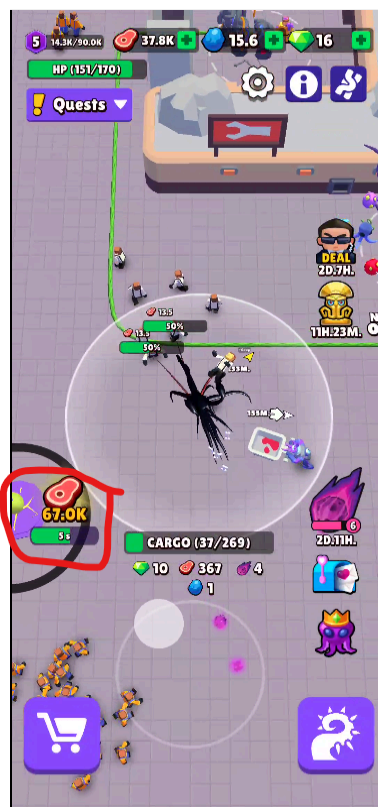
Even if some features are Ad-only or IAP-only, the game gradually switch from Ad-first to IAP-first; the best example to this is Too Expensive Evolution: at the start of the game, you can watch an ad to level up for free when evolution is too expensive, whereas later on you have to pay an important amount of HC, leading to an IAP.

When it comes to Opportunity Purchases, IAPs appear *at the beginning of the session* with time-limited bundles, whereas Ads appear *during the session* with time-limited free resources. Model ratio is respected: IAP bundles correspond to lots of resources but give a lot of time to decide, whereas Ads give very few resources but only give a few seconds to decide.

Examples:



Starting Session bundle



Ad-for-resources

Main Monetization Motivations

There are, in my opinion, 2 Main Monetization Motivations in the game:

- Progress Faster (continuous)
- Pay to Unlock Feature (specific)

I haven't included meteor monetization, as I don't know how regular are these kind of events, and how similar they are one to another

Progress Faster

As we've seen, Time is the main resource of the game and Evolution the main outcome, so Progress Faster is the main motivation.

Progress Faster gives the players what they would have gathered anyway, but gives it now. As the game progresses, the focus switch from Direct Evolution Reward (the player will evolve) to Indirect Evolution Reward (the player will get resources to evolve)

Progress Faster entry points are:

- Ads to evolve (Top entry point early game, disappears after that)
- Ads for resources (non-payers)
- Starting session bundles (most payers)
 - There is also a Starting Session Bundle of Ads to get HC
- Shop (higher spenders)
- IAPs to evolve but I guess it's a minor one, with a lower conversion rate

Progress Faster main Pay Wall is discovery of a new DNA, as when it occurs, the player usually struggles to get it.

Pay to Unlock Feature

This corresponds to specific situations where the app invites me to invest HC (more than 5) to unlock something I wouldn't get otherwise.

For the moment, I have identified 3 features, with 3 different level of pressure/visibility

First feature is Level Up Farm.

The level of pressure on this feature is quite high:

- 2 "main" quests are linked to this feature, "Level-up farm to level 2" and "Level-up farm to level 3"
- until it's done, these 2 quests are quite visible
- if you manage to forecast the result, it really produces a lot of food

The feature appears quite quickly in the game

In my own play experience, I have

- *unlocked the farm after 17 minutes*
- *discovered the first quest after 18 min*
- *discovered the second quest after 38 min*
- *maxed out level 1 after 43 min*

Second feature is New Bots.

The level of pressure is lower:

- the player needs to play a certain time with their 5 HC bots to understand how useful they are
- you need to enter the Bots Shop to be reminded of buying new bots option
- as Bots improvement applies to all the bots of the same category, it looks more beneficial after you reached a certain level of improvement, even if needs grow at a similar rate as capacities

In my own play experience (level 5) I have level 15 deliverer cargo, but only level 45 catching speed and 3 catchers cargo, as I haven't measured yet the usefulness of Catchers. For Deliverers, I only find them useful when I want to farm at a specific spot, and only when target has a high food per cargo rate

Last feature is Power Up in shop. It doesn't fully correspond to "specific situation", but still fits with "I can't have it just by playing the game"

- it is not very visible
- it looks like feature I already have access to (main evolution, black market)
- I need to "think" to realize how better this percentage upgrade is compared to the flat upgrade I can currently do in Food or Eggs

Data

Main KPIs

5 is not a lot. I have decided not to take into consideration acquisition-only KPIs, as Game Designers are rarely in charge of this part.

- First KPI would be D1, as it is the first gate to a sustainable game. I would expect 50% for a successful game.
- As I can't have access to every Dx, I would then choose D30, to check if the players we have hooked can stay in the long term. I would expect 5% to 10% for a successful game.
- For monetization, I would first focus on DARPU. It is not the most actionable indicator, but it aggregates the others (conversion, DARPPU, basket size, ecpm...). I would hope for 0.5\$ to 1\$ for a successful casual or hybrid casual game.
- Then, I would check "Today's new buyers"/"Today's buyers" ratio. If it's too low, we have an issue showing the value of the offers (people don't try them much, but when they do, they keep buying); if it's too high, the offers are deceptive or too powerful (people are ok to try them, but they never try again). A successful game is probably around 40%-50%
- Last, I would check ROAS30. It is not actionable but it's the best indicator of success:
 - above 1, a player is bringing more money than (s)he cost. Keep acquiring new users.
 - below 1, each new user is a loss of money. Except if you have a good ROAS90 and enough cash flow, increase ROAS30 or stop the game.

Case study

Situation: high drop off for players during the first session.

First, let's specify the vocabulary and the distinction between FTUE (First Time User Experience) and first session.

For this document, I'll define the FTUE as the designed minimal play scenario needed to discover the main features of the game, whereas the first session is the actual reality of what a player does during their first session (and how long they stay). In a healthy game, first session is supposed to be way longer than the initial FTUE (except if it's a hardcore game with lots of successive feature unlocks).

For this case study, I make the assumption that each FTUE step is already tracked with its own event (it is not what I have to ask for). So I already know, for example, how many / which proportion of players:

1. Used the stick to move
2. Killed a first human
3. Filled their stomach for the first time
4. Emptied their cargo for the first time
5. Launched/finished Baby steps
6. Evolved for the first time
7. Launched/finished Alien Power
8. (...and so on until: Launched/finished Big Heavies)

I also make the assumption that later on, all the Quests and Level-up are tracked.

With this information, I can imagine 3 situations of "high drop off for players during the first session":

1. High drop off at a specific moment, and this moment is in early steps
 - a. 3 hypotheses:
 - i. the player can't progress for technical reasons: I ask for tech data (freeze, crashes, ...)
 - ii. the player don't want to progress at this early stage because it's not the game they wanted: I ask for acquisition data like "which ads where used for this cohort? does drop correlate with a specific ad? which demographics were targeted? does drop correlates with specific targets?"
 - iii. the player can't progress for design reasons: they don't understand what to do, or they do what is not expected. If this happens, I don't ask for additional data. I question myself and try to find where I failed.
2. High drop off at a specific moment, later in the session ("Lots of players drop between quest [x] and quest [y]" or "Lots of players drop between level [x] and level [y])
 - a. Main question is "is it a definitive drop?"
 - i. From these droppers, which proportion does a second session within 7 days? <1%? 10%? 25%?
 - ii. For the players (all of them, not only the dropping ones) who reached [y]: how much time did they take to go from [x] to [y]? which proportion of them reached [y] in the first session?
3. High drop off all along the first session
 - a. This is most probably a design reason, usually because players are bored or struggling. I ask for progression data, and compare them with projected progression spreadsheet.

Improvements...

Alien Invasion seems to be a decent game, doing \$300K/month, when Stumble Guys (Top10 Game/Action Downloads US) does 800K/month², so I won't claim I will "fix it" in this doc. Still, there is most probably room for improvement, as in any game.

I will put aside monetization: nothing seems obviously wrong, and to identify real issues, I would need monetization data.

I will then focus on 2 game design related frictions I have experienced during the game.

They are different, but both can be phrased: "What should I do now?"

What should be my next activity?

In my journey to level 5, I have experienced very high variations on this subject; phases with very clear goals alternated with phases where I felt lost.

Level 3

First time is at level 3. Until then, the player has been driven by Quests. There was even a white arrow showing where the next Quest was. Last *straight* Quest was Big Heavies and the *continuous* Quest to max out Farm lvl1 to reach farm lvl2 is on (the player doesn't know yet that I need HC to finish the quest).

The first Quest the player encounters at level 3 is *reach farm lvl3* (I haven't even reached farm lvl2 yet) and the second *catch a hard satellite*. None of them can be my next short term

goal. Black Market is also unlocked, but rocketmen  are too random and most of the

time too fast, and humans with eggs  near lvl5 door are too though. There is 1 egg in

the down-right corner of lvl2 area , but it needs to respawn, it can't be my "main" activity.

At this moment, my farm produces 250 per minute (maxed out), as the Quest has

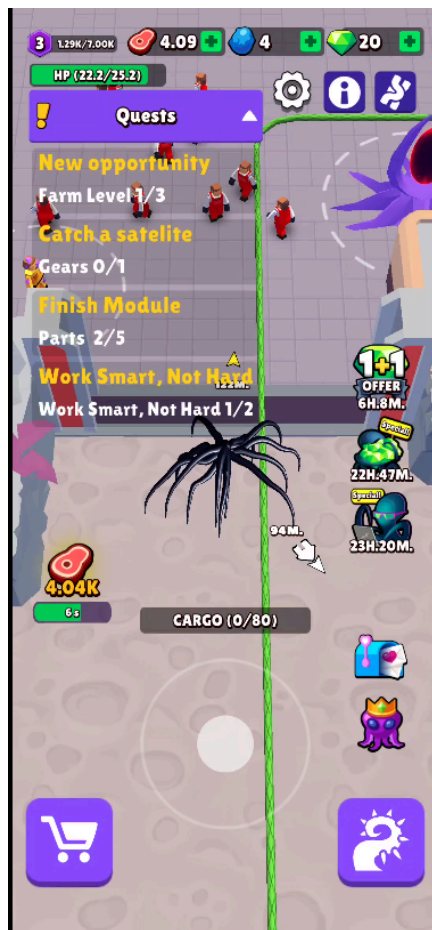
encouraged me to prioritize it. I need a lot of efforts to catch an engineer  and it only brings 218. Even richmen only bring 174. Doing a lot of effort to follow and catch a blue engineer for 504 (and 1 DNA but at this point the player doesn't know what it is for) takes me 97s, so only 25% more than the farm (lots of efforts VS doing nothing).

At this point, I *should have* upgraded my alien, but it's a vicious circle: because I have the feeling that priority should be the farm ("it brings more"+"it's what the Quest asks"), I spend my meat in the farm, not in the upgrades, increasing the "it brings more" delta.

When I max out the farm, it's a big disappointment and a "down" moment. I discover that I need 50HC to finish the Quest, and that I'm blocked. I even think "this was not the main Quest, this was a bad move", which is wrong: maxing out the farm even without leveling it up was a good move, and probably what the designers expected the player to do sooner or later.

² Source: Sensor Tower, free edition

I spend 600 (from farm) to upgrade. Satellite is now “normal”, but still difficult to catch considering the walls. I fail twice. When I manage to catch it, it gives me XP, so it brings me closer to unlocking a new area, but I still haven’t mastered this area yet.



At this point, the player has 4 Quests:

- One requires me to catch a satellite for XP. It can be done now, but as I said it doesn’t feel as a priority.
- One requires to finish the module by finishing 5 quests, most probably similar to “catch a satellite”. It is not for now, and it is not a priority either.
- One requires to spend HC. It could be a priority but it is not for now, as a non-payer needs to grind HC first (and wait several hours for the next grind opportunity).
- One requires to spend HC, fully max out the farm, spend HC again. It is definitely not for now

I understand I should set my goal to eggs. A new meaningful phase starts. But I still need 42 seconds to get 1 egg from these guys .

First shooters

My analysis of first shooters is strongly linked with my personal play experience. Still, I think a part of the players have encountered similar issues.

Shooters are often the key to next activity.

My first encounter with Shooters is Hard Astronaut, to finish Module Quest (current goal, should be my next activity). I will first die multiple times; then I will make him chase me, hurt him a lot, see him flee away and get back to 100%. Between each attempt, I upgrade my alien. Eventually, when I kill him, it's because I have over upgraded and he's now Easy. At this moment, I have the feeling I “missed” something.

My Second encounter is with level 4 guards  and DNA-guards which I need to kill to have DNA lvl2 (evolution blocked, should be my next activity). For quite a long time, I won’t try to get this DNA because I think I have to kill all the guards first to attack DNA-guards. It’s

only after several sessions that I will realize I can circle guards once I have grabbed a DNA-guard and do it long enough to kill the DNA-guard before being killed by regular ones.

Shooters have a quite complex behavior structure. From what I understood, there are phases like:

- When alien enters invisible detection range, attack
- After attack, step back. If alien is still in range, attack again.
- If alien flees, run after it, but stop running if it goes too far; it seems that this “too far range” is different from detection range (is it?)

From what I experienced:

- First encounters with Shooters, solo (astronaut) or group (lvl4 guards), is always hard and meant to fail
- Shooters from these first encounters already have the full range of complex behavior.

Agents

Currently, I have the mission to beat Agents.

I have the feeling I have understood the behavior pattern, and that it's not the issue.

Still, I am far from beating one of them.

At this point, I don't know what is the situation:

- Are Agents not my next goal? Have I misunderstood what the game wants me to do? (Maybe for some strange reasons reaching level 6 should be a higher priority, even if I struggle in level 5 area)
- Are Agents my next goal but I misunderstood how to evolve? This brings us to the next section: What should be my next improvement?

What should be my next upgrade?

Although it's obvious that upgradings must be done, some frictions appear at different steps of the decision process:

1. Characteristics can be hard to understand & What do I need now?
2. What is behind the gate?
3. I can't compare (the egg issue)

Some of these frictions only apply to main upgrade feature  , while other apply to Black Market or both

Characteristics can be hard to understand

At the beginning, when I only have cheap upgrades to do, choosing is not important: I just upgrade everything. But later on, when upgrades become quite expensive, and especially when I'm struggling with an enemy (too fast or too dangerous), I need a precise understanding of what I'm about to do.

In this situation, current descriptions of characteristics can be confusing, especially when comparing main upgrade feature and Black Market.

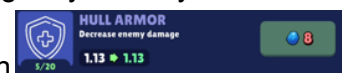


Main upgrade feature

Black Market 2*

- What is Turbo Tentacles? If it increases Move Speed, why doesn't it have the same icon?
- Is Catching Speed a damage? It seems so; but in this case, what is Strong Thread? Is it an addition to Catching Speed although it's a different icon and a different vocabulary ("Speed"+"fast" VS "Strong"+"Power")? Or is it what prevents grabbed humans from going too far away from my Alien?
- What is Fridge? It seems it increases the amount of food a human brings, but neither the icon or the name convey this idea; moreover, killing a human brings "revenue", it doesn't "cost".
- How does Hull Armor work? Currently a basic guard reduces health by 9 per hit: is it the same scale? If so, it would mean that a 8.87 damage becomes a 8.86 damage, it seems strange, as it would be a 0.11% reduction for 9 eggs. By the way, there is a

display issue with Hull Armor level5: it shows no evolution



The feature is already designed to bring some difficulties to figure out the delta resulting from the upgrade even when prices are ignored, for example:

- To catch the satellite going in walls, is it currently better to gain 40 catching speed (rough 5%), 0.5 move speed (rough 1%) or 0.3 radius (rough 10%)?

What is behind the gate?

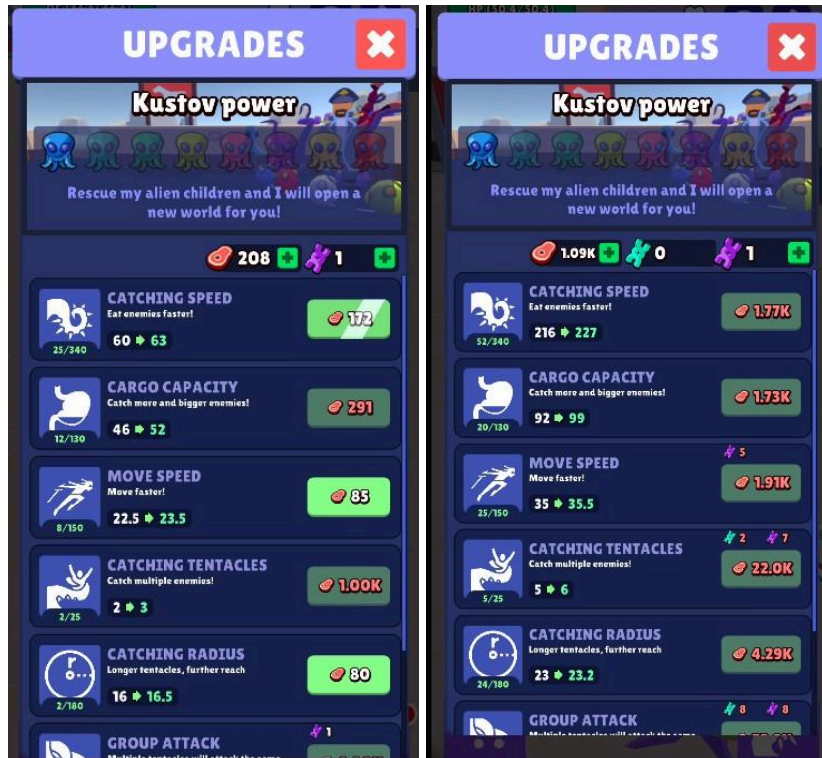
This only applies to main upgrade feature, and is linked with DNA

DNAs are scarce resources so even more than for food (or even Eggs), the player needs to know if it's worth the expense.

After spending DNA for the first few times, the player understands that DNA is a “gate” for continuous characteristics (like speed, cargo, radius, unlike catching tentacles and group attack that are not “continuous”): doing a DNA upgrade will unlock several DNA free upgrades. But it is very hard to understand **when will next DNA gate occur** (= how many free level will i have) and **what will be the cost**

A few examples below:

Level is not meaningful, as lvl25 Move Speed requires 5 DNA and lvl25 catching speed 0



Food price is not meaningful, as Cargo Capacity 30 (16.2k) requires more DNA than catching radius 43 (61k)



This full sequence is hard to get:



- Move speed 25 requires less food than cargo 25, but more DNA. Same for lvl28
- Although 28 is not a round number, Cargo 28 has a DNA gate at lvl28 (26, 27 and 29 are DNA-free)
- Lvl28 radius is DNA free
- Multiple of 10 cargo lvl30 require green DNA, but not radius 30





And we have the screenshots! Players confronted with this issue don't have this ability to analyze the sequence

I can't compare (the egg issue)

To finish the upgrade section, there is an issue with upgrading Catching Speed with eggs:

- it brings confusion to the easy message "DNA is for main upgrade feature, eggs is for Black Market"
- it makes it even more difficult to understand DNA gates
- but most of all, it requires from the player to compare 2 different screens

Improvement opportunities

Having small intermediate Quests, possibly with the same characters (once the previous quest is finished) seems feasible. These Quests would guide the player where he can be lost or make bad decisions. For example:

- "Upgrade x times" during farm quest if maxing out the farm should not be the absolute priority
- "Reach [this level] for [this characteristic]" if it's the key to defeat a difficult enemy
- "Grab x [this color] DNA" when the game detects that i should be able to start farming it
- *etc.*

Adding easier Shooters at the beginning of the game seems feasible:

- Shooter is Normal instead of Hard
- If in a group, group is 2 or 3 max, to understand what happens
- This shooter is easier to understand:
 - Option1: its detect range and chase range are visible
 - Option2: it doesn't have the full behavior. For example, it will keep chasing the Alien if it is in Catching Radius (but will not enter lvl1 area)

Clarifying the characteristics seems feasible. If still issues, maybe we need a screen (profile or other) listing all the characteristics, their definition/effect, and if it's a profile their value.

Cleaning upgrade structure to make it understandable seems difficult, as it is probably already fine tuned. Still, some marginal changes are most probably possible. A straight off (and not well-thought) example:

